

IS 8130 : 2013

(Reaffirmed 2018)

भारतीय मानक

विद्युतरोधित विद्युत केबलों एवं नम्म कॉर्डों
के लिए चालक — विशिष्टि
(दूसरा पुनरीक्षण)

Indian Standard

**CONDUCTORS FOR INSULATED ELECTRIC CABLES
AND FLEXIBLE CORDS — SPECIFICATION**
(*Second Revision*)

ICS 29.060.20

© BIS 2013

BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

October 2013

Price Group 4

Power Cables Sectional Committee, ETD 09

FOREWORD

This Indian Standard (Second Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Power Cables Sectional Committee had been approved by the Electrotechnical Division Council.

This standard was originally published in 1976 and was subsequently revised in 1984. This revision has been undertaken to bring it in line with International practices.

During the first revision of this standard, the conductors were divided into different classes, the flexibility of the conductor increasing with the class number. The number of classes of conductor were reduced to four. There are two classes of conductors for cables for fixed installations; Class 1 is for solid conductors only and Class 2 is for stranded conductors. For flexible conductors also there are two classes; as these correspond closely with Classes 5 and 6 of the previous edition, those class numbers have been retained to preserve continuity and avoid any confusion. Classes 3 and 4 were omitted, since they have had relatively little use and Classes 2 and 5 respectively were considered suitable for most of the applications for which Classes 3 and 4 had been employed.

Also during the first revision, the number of different specified maximum resistance values for different types of conductor of the same nominal cross-sectional area were reduced. The flexible aluminium conductors for welding cables were also included. Also the details about methods of various tests which were taken out from this standard were covered in a separate standard, to which reference were made at appropriate places.

Major changes in this revision include the following:

- a) Addition of a definition for nominal cross-sectional area and segmental conductor.
- b) An increase in the range of conductor sizes in Table 2.
- c) A new type of conductor has been added, that is segmental conductor.

While formulating this standard, assistance has been derived from IEC 60228 (2004) 'Conductors of insulated cables', issued by the International Electrotechnical Commission (IEC).

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

**CONDUCTORS FOR INSULATED ELECTRIC CABLES
AND FLEXIBLE CORDS — SPECIFICATION**

(Second Revision)

1 SCOPE

1.1 This standard specifies the properties and construction of copper and aluminium conductors for insulated electric cables and flexible cords.

1.2 This standard does not apply to,

- a) conductors for use in coils of machines or apparatus;
- b) conductors without insulation, for use in aerial lines;
- c) conductors for telecommunication purposes; and
- d) conductors of special design, for example, hollow-core conductors.

2 REFERENCES

The following standards contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below.

<i>IS No.</i>	<i>Title</i>
613 : 2000	Copper rods and bars for electrical purposes
1885 (Part 32) : 1993/IEC 50-461 : 1984	Electrotechnical vocabulary: Part 32 Electric cables
10810 (Part 1) : 1984	Method of test for cables: Annealing test for wires used in conductors
(Part 2) : 1984	Tensile test for aluminium wires
(Part 3) : 1984	Wrapping test for aluminium wires
(Part 4) : 1984	Persulphate test of conductor
(Part 5) : 1984	Conductor resistance test

3 TERMINOLOGY

For the purpose of this standard, the definitions given in IS 1885 (Part 32) and the following shall apply.

3.1 Metal Coated — Coated with a thin layer of suitable metal, such as tin or tin alloy.

3.2 Nominal Cross-Sectional Area — Value that identifies a particular size of conductor but is not subject to direct measurement.

3.3 Segmental Conductor — Circular conductor consisting of number of shaped or segment conductor and each separated by suitable separator. The number of segments or sector shaped conductor can be 4, 5 or 6.

NOTE — Each particular size of conductor in this standard is required to meet a maximum resistance value.

4 MATERIAL

4.1 Aluminium

The material for conductor shall consist of plain aluminium. It shall be one of the following grades with the corresponding tensile strength limits:

<i>Grade</i>	<i>Tensile Strength, N /mm²</i>
0	Up to and including 100
H2	Above 100 and up to and including 150
H4	Above 150

- a) For shaped solid conductors and the welding cable conductors, only Grade 0 aluminium shall be used;
- b) For conductors of cross-sectional area up to and including 10 mm², H2 or H4 Grade aluminium shall be used; and
- c) For the remaining conductors, aluminium of Grade 0, H2 or H4 may be used.

4.2 Copper

The conductors shall be made from high conductivity copper rods complying with IS 613. The conductors shall consist of tinned or untinned annealed copper as may be specified.

4.3 Form of Conductor

The conductors shall be solid, circular, shaped, compacted, stranded or bunched as required by the appropriate cable specification. The conductor shall be clean, reasonably uniform in size and shape, smooth and free from harmful defects.

IS 8130 : 2013

4.4 Joints in Conductor

4.4.1 Stranded Conductors

Joints shall be permitted in the individual wires of which the conductor is formed, but no joint shall be within 300 mm of any other joint within the same layer. The joints shall be made by resistance butt welding, fusion welding, cold pressure welding, electric welding, gas welding, brazing or silver soldering.

4.4.2 Solid Conductors (Aluminium Conductors only)

No joints shall be made in the finished solid conductor.

5 CLASSIFICATION

5.1 The conductors have been divided into four classes as follows :

- | | |
|-----------------------------------|-----------------|
| a) Cables for fixed installations | Classes 1 and 2 |
| b) The flexibles | Classes 5 and 6 |
| c) Copper welding cables | Class 6 |

5.2 Aluminium conductors for welding cables have not been assigned any class number.

6 CONSTRUCTION

6.1 Solid Conductor (Class 1)

6.1.1 The conductor shall consist of single wire of plain or tinned annealed copper or plain aluminium in accordance with Table 1.

6.1.2 Solid copper conductor shall be of circular cross-section.

6.1.3 Solid aluminium conductor of sizes from 10 mm² up to and including 35 mm² shall be of circular cross-section. Sizes 35 mm² and above may be of either circular or shaped cross-section.

6.2 Stranded Circular Non-compacted Conductors (Class 2)

6.2.1 Conductor shall consist of plain or tinned annealed copper or plain aluminium.

6.2.2 The wires in the conductor shall have the same nominal diameter before stranding.

6.2.3 The number of wires in the conductor shall be not less than the appropriate minimum number gives in Table 2.

6.3 Stranded Compacted Circular Conductors and Shaped Conductors (Class 2)

6.3.1 Conductor shall consist of plain or tinned annealed copper or plain aluminium.

6.3.2 The ratio of the diameters of two wires before stranding in the same conductor shall not exceed 2.

6.3.3 The number of wires in the conductor shall be not less than appropriate minimum number given in Table 2.

6.4 Flexible Conductors (Classes 5 and 6)

6.4.1 Conductor shall consist of plain or tinned annealed copper.

6.4.2 The wires in the conductor shall have same nominal diameter before bunching.

6.4.3 The diameter of the wires in any conductor shall not exceed the appropriate maximum value given in Table 3 for Class 5 and Table 4 for Class 6 conductors.

6.5 Flexible Aluminium Conductors for Welding Cables

6.5.1 Conductor shall consist of plain aluminium.

6.5.2 The wires in the conductor shall have same nominal diameter before stranding.

6.5.3 The diameter of wires in any conductor shall not exceed the appropriate maximum value given in Table 5.

7 TESTS

7.1 Tests for Copper Conductors

7.1.1 Persulphate Test (for Tinned Copper only)

This test shall be carried out in accordance with Method B given in IS 10810 (Part 4). The mass of copper dissolved shall not exceed the appropriate value given below:

Wire Diameter mm	Permissible Mass of Copper Dissolved, Max g/m ²
Up to and including 0.41	5
Above 0.41	3

7.1.2 Annealing Test

7.1.2.1 Before standing

This test shall be carried out in accordance with IS 10810 (Part 1). The elongation shall be not less than the appropriate value given below:

Wire Diameter		Elongation, Min
Over mm	Up to and Including mm	Percent
—	0.21	9.0
0.21	0.41	13.5
0.41	1.36	18.0
1.36	—	22.5

IS 8130 : 2013

7.1.2.2 After stranding

Under consideration.

7.2 Tests for Aluminium Conductors

7.2.1 Tensile Test

This test shall be carried out in accordance with IS 10810 (Part 2). The sample shall meet the following requirements:

- a) *Before stranding* — The tensile strength of any of the wires shall comply with the values given in **3.1**.
- b) *After stranding* — The tensile strength of any of the wires shall be not less than 95 percent of the minimum values given in **3.1**.

NOTE — This test for wires taken from stranded conductors is not applicable in the case of compacted circular conductors or shaped conductors.

7.2.2 Wrapping Test

This test shall be done in accordance with IS 10810 (Part 3). The criteria for passing is that the wire shall not break.

NOTE — This test is not applicable to shaped solid conductors, wires taken from stranded conductors in case of compacted circular or shaped conductors and wires used for welding cable conductors.

7.2.3 Annealing Test

This test is applicable to shaped solid conductor and wires of conductor (before stranding) for welding cables only. The test shall be carried out in accordance with IS 10810 (Part 1). Shaped solid conductors shall have a minimum of 25 percent and wires of conductor

for welding cables shall have a minimum of 12 percent elongation.

7.3 Resistance Test (for Both Copper and Aluminium)

This test shall be done on the finished conductors in accordance with IS 10810 (Part 5). The dc resistance of the conductor shall be measured at room temperature and corrected to 20°C by means of the appropriate factors given in Table 6.

7.3.1 The corrected dc resistance shall not exceed the values given in the appropriate Table 1 to Table 5.

8 PACKING AND MARKING

8.1 The conductor shall either be wound on reels or drums or supplied in coils packed and labelled. The label shall contain the name of the manufacturer, trade name, if any; size and length of the conductor.

NOTE — The requirements of packing and marking are not applicable when the processing of the conductor forms part of the manufacturing of complete cable.

8.2 BIS Certification Marking

The conductor (reel, drum or label) may also be marked with the Standard Mark.

8.2.1 The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act, 1986* and the Rules and Regulations made thereunder. The details of conditions under which a license for the use of the Standard Mark may be granted to the manufacturers or producers may be obtained from the Bureau of Indian Standards.

IS 8130 : 2013

Table 1 Solid Conductors for Single Core and Multicore Cables (Class 1)
(Clauses 6.1.1 and 7.3.1)

Sl No.	Nominal Cross-Sectional Area mm ²	Maximum Resistance of Conductor at 20° C		
		Circular Copper Conductors		Aluminium Conductors Circular or Shaped
		Plain Ω/km	Tinned Ω/km	Ω/km
(1)	(2)	(3)	(4)	(5)
i)	0.5	36.0	36.7	—
ii)	0.75	24.5	24.8	—
iii)	1	18.1	18.2	—
iv)	1.5	12.1	12.2	—
v)	2.5	7.41	7.56	—
vi)	4	4.61	4.70	—
vii)	6	3.08	3.11	—
viii)	10	1.83	1.84	3.08 ²⁾
ix)	16	1.15	1.16	1.91 ²⁾
x)	25	0.727 ¹⁾	—	1.20 ²⁾
xi)	35	0.524 ¹⁾	—	0.868 ²⁾
xii)	50	0.387 ¹⁾	—	0.641
xiii)	70	0.268 ¹⁾	—	0.443
xiv)	95	0.193 ¹⁾	—	0.320 ³⁾
xv)	120	0.153 ¹⁾	—	0.253 ³⁾
xvi)	150	0.124 ¹⁾	—	0.206 ³⁾
xvii)	185	—	—	0.164 ³⁾
xviii)	240	—	—	0.125 ³⁾
xix)	300	—	—	0.100 ³⁾

¹⁾ The solid copper conductors having nominal cross-sectional areas of 25 mm² and above are intended for particular types of cables only and not for general purpose.

²⁾ Aluminium conductors 10 mm² to 35 mm² circular only.

³⁾ For single core cables, four sectoral shaped conductor may be assembled into a single circular conductor.

Table 2 Stranded Conductors for Single Core and Multicore Cables (Class 2)
(Clauses 6.2.3, 6.3.3 and 7.3.1)

Sl No.	Nominal Cross- Sectional Area	Minimum Number of Wires in the Conductor				Maximum Resistance of Conductor at 20°C		
		Circular Conductor (Non-Compacted)		Circular Compacted or Shaped Conductor		Copper Conductor		Aluminium Conductor
		Cu	Al	Cu	Al	Plain Wires Ω/km	Tinned Wires Ω/km	Ω/km
(1)	mm ² (2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	1	3	—	—	—	18.1	18.2	—
ii)	1.5	3	3	—	—	12.1	12.2	18.1
iii)	2.5	3	3	—	—	7.41	7.56	12.1
iv)	4	7	3	—	—	4.61	4.70	7.41
v)	6	7	3	—	—	3.08	3.11	4.61
vi)	10	7	7	6	—	1.83	1.84	3.08
vii)	16	7	7	6	6	1.15	1.16	1.91
viii)	25	7	7	6	6	0.727	0.734	1.20
ix)	35	7	7	6	6	0.524	0.529	0.868
x)	50	19	19	6	6	0.387	0.391	0.641
xi)	70	19	19	12	12	0.268	0.270	0.443
xii)	95	19	19	15	15	0.193	0.195	0.320
xiii)	120	37	37	18	15	0.153	0.154	0.253
xiv)	150	37	37	18	15	0.124	0.126	0.206
xv)	185	37	37	30	30	0.099 1	0.100	0.164
xvi)	240	61	37	34	30	0.075 4	0.076 2	0.125
xvii)	300	61	61	34	30	0.060 1	0.060 7	0.100
xviii)	400	61	61	53	53	0.047 0	0.047 5	0.077 8
xix)	500	61	61	53	53	0.036 6	0.036 9	0.060 5
xx)	630	91	91	53	53	0.028 3	0.028 6	0.046 9
xxi)	800 ¹⁾	91	91	53	53	0.022 1	0.022 4	0.036 7
xxii)	1 000 ¹⁾	91	91	53	53	0.017 6	0.017 7	0.029 1
xxiii)	1 200 ²⁾	—	³⁾	—	—	0.015 1	0.015 1	0.024 7
xxiv)	1 400 ²⁾	—	³⁾	—	—	0.012 9	0.012 9	0.021 2
xxv)	1 600 ²⁾	—	³⁾	—	—	0.011 3	0.011 3	0.018 6
xxvi)	1 800 ²⁾	—	³⁾	—	—	0.010 1	0.010 1	0.016 5
xxvii)	2 000 ²⁾	—	³⁾	—	—	0.009 0	0.009 0	0.014 9
xxviii)	2 500 ²⁾	—	³⁾	—	—	0.007 2	0.007 2	0.012 7

¹⁾ These are segmental conductors.

²⁾ Can be either stranded or segmental.

³⁾ The minimum number of wires for these sizes is not specified. These sizes may be constructed from 4, 5 or 6 equal segments (Milliken).

NOTE — In case 800 and 1 000 square mm if constructed as per segment construction then the minimum number of wires is not specified. However the number of segments can be 4, 5 or 6.

IS 8130 : 2013

Table 3 Flexible Copper Conductors for Single Core and Multicore Cables (Class 5)
(Clauses 6.4.3 and 7.3.1)

Sl No.	Nominal Cross- Sectional Area	Maximum Diameter of Wires in Conductor	Maximum Resistance of Conductor at 20° C	
			Plain Wires	Tinned Wires
(1)	mm ² (2)	mm (3)	Ω/km (4)	Ω/km (5)
i)	0.5	0.21	39.0	40.1
ii)	0.75	0.21	26.0	26.7
iii)	1	0.21	19.5	20.0
iv)	1.5	0.26	13.3	13.7
v)	2.5	0.26	7.98	8.21
vi)	4	0.31	4.95	5.09
vii)	6	0.31	3.30	3.39
viii)	10	0.41	1.91	1.95
ix)	16	0.41	1.21	1.24
x)	25	0.41	0.780	0.795
xi)	35	0.41	0.554	0.565
xii)	50	0.41	0.386	0.393
xiii)	70	0.51	0.272	0.277
xiv)	95	0.51	0.206	0.210
xv)	120	0.51	0.161	0.164
xvi)	150	0.51	0.129	0.132
xvii)	185	0.51	0.106	0.108
xviii)	240	0.51	0.080 1	0.081 7
xix)	300	0.51	0.064 1	0.065 4
xx)	400	0.51	0.048 6	0.049 5
xxi)	500	0.61	0.038 4	0.039 1
xxii)	630	0.61	0.028 7	0.029 2

Table 4 Flexible Copper Conductors for Single Core and Multicore Cables (Class 6)
(Clauses 6.4.3 and 7.3.1)

Sl No.	Nominal Cross- Sectional Area	Maximum Diameter of Wires in Conductor	Maximum Resistance of Conductor at 20° C	
			Plain Wires	Tinned Wires
(1)	mm ² (2)	mm (3)	Ω/km (4)	Ω/km (5)
i)	0.5	0.16	39.0	40.1
ii)	0.75	0.16	26.0	26.7
iii)	1	0.16	19.5	20.0
iv)	1.5	0.16	13.3	13.7
v)	2.5	0.16	7.98	8.21
vi)	4	0.16	4.95	5.09
vii)	6	0.21	3.30	3.39
viii)	10	0.21	1.91	1.95
ix)	16	0.21	1.21	1.24
x)	25	0.21	0.780	0.795
xi)	35	0.21	0.554	0.565
xii)	50	0.31	0.386	0.393
xiii)	70	0.31	0.272	0.277
xiv)	95	0.31	0.206	0.210
xv)	120	0.31	0.161	0.164
xvi)	150	0.31	0.129	0.132
xvii)	185	0.41	0.106	0.108
xviii)	240	0.41	0.080 1	0.081 7
xix)	300	0.41	0.064 1	0.065 4

Table 5 Flexible Aluminium Conductors for Welding Cables
(Clauses 6.5.3 and 7.3.1)

Sl No.	Nominal Cross-Sectional Area mm ²	Maximum Diameter of Wires in Conductor mm	Maximum Resistance of Conductor at 20°C Ω/km
(1)	(2)	(3)	(4)
i)	25	0.31	1.23
ii)	35	0.31	0.901
iii)	50	0.31	0.634
iv)	70	0.31	0.445
v)	95	0.31	0.334
vi)	120	0.31	0.256

Table 6 Temperature Correction Factors k_t for Conductor Resistance to Correct the Measured Resistance at t °C to 20°C
(Clause 7.3)

Temperature of Conductor at time of Measurement t °C	Correction Factor k_t	Temperature of Conductor at Time of Measurement t °C	Correction Factor k_t
(1)	(2)	(1)	(2)
5	1.064	28	0.969
6	1.059	29	0.965
7	1.055	30	0.962
8	1.050	31	0.958
9	1.046	32	0.954
10	1.042	33	0.951
11	1.037	34	0.947
12	1.033	35	0.943
13	1.029	36	0.940
14	1.025	37	0.936
15	1.020	38	0.933
16	1.016	39	0.929
17	1.012	40	0.926
18	1.008	41	0.923
19	1.004	42	0.919
20	1.000	43	0.916
21	0.996	44	0.912
22	0.992	45	0.909
23	0.988	46	0.906
24	0.984	47	0.903
25	0.980	48	0.899
26	0.977	49	0.896
27	0.973	50	0.893

k_t is based on resistance temperature co-efficient of 0.004 per °C at 20°C using following formula :

$$k_t = \frac{1}{1 + 0.004(t - 20)} = \frac{250}{230 + t}$$

Bureau of Indian Standards

BIS is a statutory institution established under the *Bureau of Indian Standards Act, 1986* to promote harmonious development of the activities of standardization, marking and quality certification of goods and attending to connected matters in the country.

Copyright

BIS has the copyright of all its publications. No part of these publications may be reproduced in any form without the prior permission in writing of BIS. This does not preclude the free use, in the course of implementing the standard, of necessary details, such as symbols and sizes, type or grade designations. Enquiries relating to copyright be addressed to the Director (Publications), BIS.

Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards : Monthly Additions'.

This Indian Standard has been developed from Doc No.: ETD 09 (6339).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002

Telephones : 2323 0131, 2323 3375, 2323 9402

Website: www.bis.org.in

Regional Offices:

Telephones

Central	: Manak Bhavan, 9 Bahadur Shah Zafar Marg NEW DELHI 110002	{ 2323 7617 2323 3841
Eastern	: 1/14 C.I.T. Scheme VII M, V. I. P. Road, Kankurgachi KOLKATA 700054	{ 2337 8499, 2337 8561 2337 8626, 2337 9120
Northern	: SCO 335-336, Sector 34-A, CHANDIGARH 160022	{ 260 3843 260 9285
Southern	: C.I.T. Campus, IV Cross Road, CHENNAI 600113	{ 2254 1216, 2254 1442 2254 2519, 2254 2315
Western	: Manakalaya, E9 MIDC, Marol, Andheri (East) MUMBAI 400093	{ 2832 9295, 2832 7858 2832 7891, 2832 7892

Branches: AHMEDABAD. BANGALORE. BHOPAL. BHUBANESHWAR. COIMBATORE. DEHRADUN. FARIDABAD. GHAZIABAD. GUWAHATI. HYDERABAD. JAIPUR. KANPUR. LUCKNOW. NAGPUR. PARWANOO. PATNA. PUNE. RAJKOT. THIRUVANANTHAPURAM. VISAKHAPATNAM.